

1) Continuous part of an array is called subarray

2) A single element is a subarray

3) Entire array is a subarray

4) Empty cannot be a subarray.

* To uniquely define a subarray:-

Subarray [start end]

Subarray [i j] $\rightarrow$ length = $j - i + 1$
 $(i \leq j)$

* Count of subarray:-

[4 2 10 3 12 -2 15]
0 1 2 3 4 5 6

$$= \frac{N(N+1)}{2} = \frac{7(7+1)}{2} = 28$$

* Print values of subarray:-

Tc: $O(N)$

Sc: $O(1)$

* Find Sum in Subarray:-

$$Tc: O(N)$$

$$Sc: O(1)$$

* Print All Subarray:-

$$Tc: O(N^3)$$

$$Sc: O(1)$$

* Print Sum of all Subarray:-

$$Tc: O(N^2)$$

$$Sc: O(1)$$

optimisation:-

$$\text{sum}[L, R] = \begin{cases} \text{if } L=0 & pf[R] \\ \text{else} & pf[R] - pf[L-1] \end{cases}$$

$$Tc: O(N^2)$$

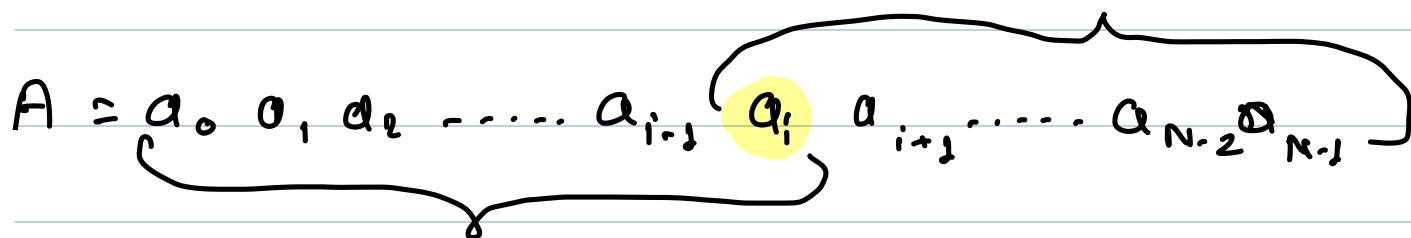
$$Sc: O(N)$$

* Print all Subarray Sums using Carry forward:-

$$Tc: O(N^2), Sc: O(1)$$

* Sum of all subArray sum:-

Given arr[N], in how many subarrays index i is present?



start $[0, i] \Rightarrow (i+1)$ elements

end $[i, N-1] \Rightarrow (N-i)$ elements

In How many subarray is i^{th} elements present.

$$(i+1) * (N-i)$$